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Designing Your Research Study

Designing Your Research Study

Research Questions and Hypotheses

The first step to designing a study is to determine your research question. A research question identifies your main problem or question you are aiming to answer and signals what type of study you need to conduct. For instance, your question may be “*What elements of a TV ad are associated most with experiencing positive emotions?*”

A hypothesis is a testable answer to your research question. It includes the specific variables that are going to be tested. A hypothesis to the above question may be “*Narrative and auditory elements have the highest influence on brand liking and acceptance.*” In scientific studies, hypotheses are generated *before* designing your study and collecting any data. This allows you as the researcher to generate a theory behind the research that already predicts potential answers to the research question. Running the study, therefore, serves to test that theory’s expectations. Moreover, hypothesizing prior to data collection prevents interpreting the results in line with your expectations, or *confirmation bias*¹. In some commercial settings, studies can also be done in an exploratory fashion, without having fixed hypotheses. Still, it is a good idea to watch out for confirmation bias!

Variables

The variables are the objects that will be manipulated and measured during the study. In the above example, TV ad elements such as *narrative*, *duration*, and *color contrasts*, are the variables that change across TV ads. The changed or manipulated variables are the *independent variables*. Further, experiencing positive emotions is the *dependent variable* that we want to measure in response to changing video elements.

Variables are also categorized as their data being either *discrete* or *continuous*, depending on how they are measured. For instance, “number of children” is a discrete variable, since the answer can only be a discrete number such as zero, one, or two or more, however, it cannot be a number between these values. Discrete variables, in general, can be *counted*, whereas a continuous variable can have any numeric value. For instance, “reaction time” is a continuous variable that is uncountable and has to be measured on a continuous scale.

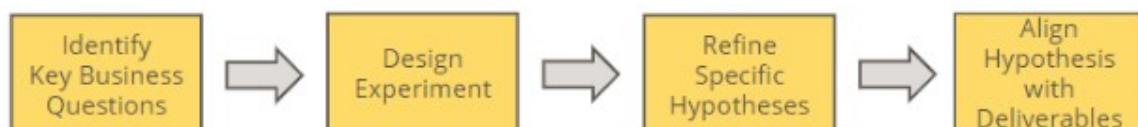
Whether your independent and dependent variables are discrete or continuous has huge implications for designing the most appropriate study for your research question, and deciding on what statistical test to use.

Examples of good and bad research questions

Good	Qualities	Bad	Qualities
What elements of a TV ad are associated most with experiencing positive emotions?	- Identifies the variables - Quantifiable and measurable - Specific materials are indicated - Specific populations are indicated - What/How question	What kind of TV ad is the best?	- Opinion based - Too general - Missing material specifications - Missing population specifications - Have Yes/No answer - Too simple to test
What is the impact of music on how consumers respond to my creative ?		Does music impact consumer response to my creative ?	
How does the amount of interactivity in an ad impact the emotional response that users have to that ad?		Is it good to have interactivity in an ad?	
How does the frequency of instructional elements appearing on-screen during a tutorial affect how players respond to the tutorial?		Do instructional elements improve the response of users to tutorials?	

To carry out an effective study design, follow the below scheme and key considerations.

Study Design



1. First, identify the specific business question and how biosensor research can help to answer this question. To identify the business question, first:

- Determine the core of what your target client is trying to answer and think about the business impact of what they are asking. In other words, dive deeper and avoid taking the objective from a research brief at face value.
- Think about what can be controlled, such as the number of stimuli or variables, or the environment. High-quality research is critical, but it is equally as important to recognize that clients desire ecological validity for their research results. Be sure to balance your study with real-world implications.

- Secondly, design your experiment. This process first involves deciding what kind of stimuli is needed to answer the business question. Start by identifying key words in the client's business questions that may point your research in the direction of certain methodologies. Additionally, determine the sample size needed. Consider the below questions:
 - What sample size (number of participants) do I need?
 1. The major determining factors for this question are the expected size of the effect for which you are testing, and the precision and validity of your measurement tools and experimental design.
 - How much content can I expose someone to?

Two keys aspects must be considered here:

 1. Length of respondent sessions (when do you worry about fatigue)?
 2. Similarity of different stimuli to one another
 - What are the appropriate modalities?
 - What is the right balance between controlling the experience, and being "true" to real life?
 - What does the client want, versus what can they afford?
- The next step is to refine your hypothesis. Once the study design is confirmed, you can solidify more specific hypotheses. Take into account what kind of comparisons are being made in your study. By setting hypotheses and expectations *before the study*, you are ensuring that researchers or clients are not going on a "fishing expedition". Try to avoid conducting studies "just to see what we can see". Biometric measures are granular enough that you can almost always find some result, but without a specific business question, that result may just be found due to random chance. See the section on multiple comparisons problem.
- Finally, align hypotheses with deliverables. Decide on the specific metrics you will use to answer each of your questions and which sensors are best to obtain the results you are looking for.

Here you have an example of good and less-optimal study design. In both cases we are using the same business question:

Example of a **Good** Study Design

What is the impact of music on how consumers respond to my creative?

Good Design

- 6 versions of same **visual** creative, with different music in each
- Each piece of creative tested with a separate group of respondents, to prevent priming effects
- Ideally, music is bucketed into categories – e.g., 3 examples of “serious” music and 3 examples of “light/fun” music

Why is this a good design?

1. 6 examples is typically sufficient to control for creative effects – e.g., avoiding having results skewed by one particularly good or particularly bad piece of music
2. By holding the visual creative constant, we isolate the effect of **just music**, allowing us to draw conclusions about its impact
3. Grouping the types of music into higher-order categories – “serious” vs. “fun”, for example – allowing for generalized insights about genres of music

Example of a **Less-Optimal** Study Design

What is the impact of music on how consumers respond to my creative?

Less-Optimal Design

- 6 different ads, with different variations of music, visual style, and story
- Respondents are exposed to multiple pieces of creative over the course of the experience
- Each piece of music is from a different genre

Why does this design need improvement?

1. Changing multiple variables within the same test makes it impossible to isolate the effect of any one variable: It would not be possible to attribute responses to music alone, because differences could come from changes in visuals, etc.
2. Because all of the pieces of creative are totally different, exposing respondents to multiple creatives is not problematic by itself – however, because of #1 above, it will be difficult to attribute results across respondents to any one specific variable
3. Having each piece of music be from a different genre does not appropriately control for genre/creative effects, and makes it impossible to draw generalized conclusions about types of music and their effectiveness

Pretest/Post-test Design

Pretest/post-test design is one of the most common research designs. In this kind of study, the researcher takes measurement *before* and *after* an intervention or exposure to a stimulus. For instance, emotional measurements can be taken before watching an emotionally arousing video and after. The pre- and post-measures can then be compared to see if the stimulus resulted in any increase or decrease in the emotion measures.

Between & Within subjects designs

There are two main categories of research designs in terms of the allocation of participants to the different experimental conditions. If you are testing the usability of three different websites, for instance, will your participants test all three websites, or will you have different groups of participants testing only one kind of website? The former condition is between-subjects design, in which the participants are assigned to only one of the independent variable's levels (e.g. kind of website). The latter is the within-subjects design, in which all participants join all conditions of the study (e.g. they use all three websites in different orders).

In between-subjects designs, the study sessions are shorter in duration. However, a higher sample size is needed, since every respondent will join only one condition.

Within-subjects studies typically cost less, since fewer participants are necessary to run the study. Also, this design has less error from confounding variables such as personality and mood, with each individual serving as their control. However, it is important to keep in mind that participants may become too tired and bored if you are testing too many conditions. This exposure fatigue may alter the results of your study.

In more complex studies, a mixed design can be used implementing both between and within-subjects designs for different factors in a study. For instance, the type of website can be a within-subject factor whereas the type of device used when browsing these websites (e.g. mobile versus laptop) can be a between-subjects factor. The below table demonstrates that in such a design, the device type is the between-subjects factor, whereas the website type is the within-subjects factor. So, one participant will be randomly assigned to either the laptop or phone condition, and he or she will test all three websites in one device.

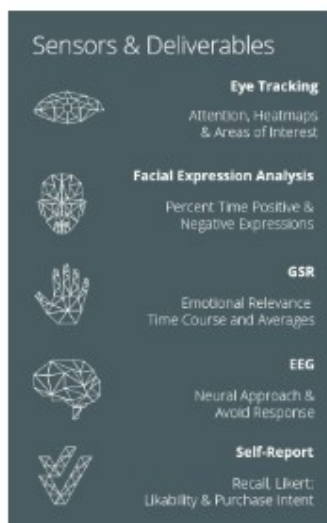
Between-subjects factor	
Laptop	Mobile Phone
Website 1	Website 1
Website 2	Website 2
Website 3	Website 3

Within-subjects factor

One critical consideration when creating your study is randomization. In most testing scenarios, the order of materials must be randomized because a participant testing multiple conditions might come to familiarize themselves with the experiment, stimuli, or the task, influencing the results based on the order of presentation. Moreover, fatigue effects might occur, causing the participant to become more distracted and tired towards the end of a session. If the order of stimuli is randomized for each participant, these effects would cancel each other out when results are aggregated. This procedure is called *counterbalancing*.

Standard Methodologies

- Video Content Testing: Used for TV/Digital Ads, long-form videos and simulated environments. The most common sample business questions are used to:
 - Identify videos, or moments within videos, that connect emotionally with consumers
 - Evaluate video creative iteratively at multiple stages (animatic, unfinished creative, finished creative)
 - Understand which, of multiple, creatives may perform best in-market
 - Determine the impact of creative elements (music, characters, etc.)
 - Analyze the emotional journey viewers experience, at an aggregate level, on a moment to moment basis throughout the creative



- **Screen-Based Static Image Testing:** Used for Print ads, packaging and shelf displays. The most common sample business questions are used to:
 - Determine which imagery to bring to market, and which designs may be eliminated
 - Evaluate how well the stimulus is noticed in a specific context, and what stimuli may go ignored or missed
 - Identify what stimulus elements (e.g., key messaging) get noticed by consumers
 - Understand what stimulus elements connect with consumers emotionally, and which may not

Static Image Sample Report Download



- **Sensory Testing:** Used for fragrances and flavors. The most common sample business questions are used to:
 - Identify candidate formulations for use in future products
 - Evaluate responses to new formulations relative to previous, or to competitor, formulations
 - Understand responses to individual formulation ingredients
 - Find formulations that may elicit specific profiles (e.g., relaxation or excitement)

Sensory Sample Report Download



- **UX Testing:** Used for web-based desktop or mobile UX and mobile apps. The most common sample business questions are used to:
 - Reveal what elements on a web page get noticed, and which get missed
 - Understand what parts of the user experience elicit an emotional response from consumers
 - Identify potential areas to optimize page “flow”, from behavior and visual attention
 - Determine the natural navigation path of users through the web page or content
 - Determine what pages lead users to express facial frustration or confusion

UX Sample Report Download



Determining Appropriate Modalities

What is a “modality”?

Modality is a term used in behavioral research referring usually to one way or one aspect of measurement. For example, we can measure attention using the modality of eye-tracking. Similarly, we can measure emotion using EEG and facial expression analysis modalities. That way, we get at the constructs of interest from multiple dimensions, making our findings more robust. In other words, each biosensor is considered to be one measurement modality.

How to determine which modalities to use:

It is recommended to get acquainted with the combination of biosensor modalities for different research application areas by reading our [sample reports](#) and standard methodologies above. For example, it typically does not make sense to use facial expression analysis when testing static images because facial expressions are mostly observed either in social settings when individuals are more concerned with signaling emotions to one another or in response to stronger and more dynamic emotional content. Measuring facial expressions while consumers are watching static content may not be useful for gathering insights about the content. Instead, FEA is recommended as a modality to assess something like the response to arousing and dynamic videos to assess the valence and activation levels of emotions elicited. Alternative modalities such as ET and EEG can provide more granular, useful information in the context of static images.

Eye-tracking is always a recommended modality when assessing visual attention. When testing an image, a video, an environment, a website, or any visual content, eye-tracking can provide insights on which parts of the stimulus are noticed and what is missed. Visual attention may also be a proxy of liking, since people tend to allocate visual attention to arousing and approach-eliciting stimuli. To that end, it is possible to combine GSR and EEG with ET to study the moments where attention is highly fixed on a particular point and determine if arousal and approach responses are also high. Then, it is possible to conclude that a specific point in a dynamic stimulus caused a strong emotional, or potential liking response.

Sampling & Random Assignment

Sampling is the process by which participants are chosen and recruited. How many people will participate in your study? What are their characteristics, such as age, group, gender, income level, and geographical region? Are there any other characteristics of your target market that you would like to recruit specifically for? The number of participants in your study is your *sample size*. Sample size is determined by the type of study you will conduct. Generally, we recommend 20-60 people in each study condition for adequate statistical power. This number depends on many things such as what sensors you use, what kind of analysis you wish to conduct, and the desired effect size².

After sampling and recruiting your participants, it is a crucial step to randomly assign participants to one of the experimental groups. To avoid any bias, each participant must have an equal chance to participate in one of the conditions. Random assignments can be easily automated in iMotions using block design procedures. See our [Help Center article](#) for more technical information about preparing a block design in iMotions. Random assignment can also be done manually using random number generator resources such as [randomizer.org](https://www.randomizer.org).

Instructing Participants

Instructing participants clearly increases the validity and accuracy of research data. It should be communicated clearly to the participants what to expect in the experience, and what they will be doing during a study session.

It is recommended to prepare written instructions in advance of your study to ensure all necessary information is communicated, and that this communication is relatively uniform for all participants. Participants can also read the instructions while you are setting up the equipment. Written instructions can also be implemented on your stimuli set on iMotions as a survey slide. Instructing participants is critical to ensure smooth data collection as well as quality data. An example instruction slide for a video ad test may be:

Thank you for your participation. Today you will be viewing a set of visual stimuli and then answering questions pertaining to that stimuli. Please do not obstruct your face during the session and keep your hands relaxed. The visual stimuli will advance automatically. When you are ready to begin, click Next.

For more sample scripts, click [here](#).

Informed Consent

All participants must be informed and understand the most relevant information about the study they are going to be part of before they begin. Informed consent notifies participants about the general purpose of the research, modalities used, and any risk involved, without giving away any priming or critical information about the results or outcome of the study that may introduce bias. Participants also receive information about how their personal data is going to be stored and processed, the legal basis, and their rights as respondents. They are also notified that the obtained results will be anonymized and, therefore, would not be linked to them as a person.

Informed consent is a matter of ethics, and the importance cannot be understated. In some countries, informed consent is actually a legal obligation. Ethical research practice is not only important for your reputation as a researcher, but also for avoiding any legal complications down the road.

All of the respondents must also know that it is *voluntary* to participate in the study. At any time, they can withdraw their consent orally, in written form, or in any other clearly understandable way to terminate their participation in the study. Withdrawing from the study will not affect their participation in other current or future studies or any other rights to justify this decision.

In order to confirm consent in the research, participants are instructed to sign a consent document where they declare that they have received written and oral information about the study, and they know enough about the purpose and methods of this study to be able to agree to participate in it. Please consult your legal departments to design a consent form that properly protects your participants, and considers all necessary legal implications.

¹ Confirmation bias is the tendency to perceive information that is in line with your prior beliefs and expectations, while ignoring or not accepting information that goes against your beliefs and expectations.

² Effect size is a value that quantifies the size or magnitude of the difference between two group averages.

